特集論文／医用イメージングにおける画像再構成の基礎と現状

Overview of Compressed Sensing Approach to CT Image Reconstruction

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Abstract

This paper provides an overview of the compressed sensing approach and its contribution in the development of reconstruction algorithms in x-ray computed tomography. We briefly review the recent advances in the topic and refer to the related applications. Finally, we discuss the challenges related to the development of the compressed sensing based reconstruction algorithms and possibilities to be used in clinical scanners.

Key words: Compressed sensing, Image reconstruction, Computed tomography

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1. Introduction

Computed tomography (CT) is one of the leading imaging modalities with several applications in medicine, biology and industry. Cross sectional images/volumes are obtained through a reconstruction from projection data, which computed through radiating the object with x-ray photons and measure the non-attenuated portion using detectors located on the opposite side of the x-ray source. By rotating the scanning system or the target object, several projection views are acquired. The fractional between the transmitted and the measured photons along specific ray directions is used to reconstruct the CT image. The reconstructed image represents the attenuation distribution of x-ray photons inside the scanned object. The sampling rate, source trajectory and other scanner geometric properties are important factors for exact and stable reconstruction. For example, if the number of projection views is small (e.g. 10 views), the reconstructed image with conventional analytical methods, such as the filtered back-projection (FBP) algorithm, is known to suffer from streak/aliasing artifacts. This phenomenon is acceptable within the framework of Shannon/Nyquist sampling theorem, which states that the sampling rate should be at least double the highest frequency of nonzero magnitude.

Recently, the theory of compressed sensing (or compressive sampling) provides an interesting framework where the image can be reconstructed from far fewer data measurements than it is conventionally required [1, 2]. The main idea is based on presenting the image object into some domain where most of the coefficients are zeros or close to zero. This sparse formulation is much easier to solve through ℓ_1 norm minimization when the number of data measurement is small. The reason behind this is that, ℓ_1 norm minimization is more powerful compared to conventional methods in finding the sparse solution, especially when the data measurement is undersampled. The CS framework became a useful tool in many problems suffers from data limitation in several fields such as signal processing, machine learning, computational biology, astronomy, communications and medical imaging. Although the term compressed sensing was not being used before the work in [1, 2], The same idea was introduced earlier for CT reconstruction of sparse objects such as blood vessels [3].

Image reconstruction from limited projection data is a conventional problem in CT with several potential clinical applications. It is important for the reduction of patient dose, improvement of image quality and other similar

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requirements. This clarifies the reason behind the extensive use of CS framework in developing effective image reconstruction algorithms from limited projection data. This paper mainly aims to provide an overview of the CS approach and discuss the recent advances in CT image reconstruction.

2. Compressed Sensing Problem in CT

In general, the term CS refers to an approach that is used to find the sparse solutions to an underdetermined linear system. Let $\boldsymbol{x} = (x_1, \dots, x_n)$ is the image vector representing the attenuation coefficients of the imaged object, $\boldsymbol{y} = (y_1, \dots, y_m)$ is the measured projection data (line integrals) and $A = \{a_{ij}\}$ is the $m \times n$ system matrix. The image reconstruction problem can be simply formulated as the following liner system:

$$A\boldsymbol{x} = \boldsymbol{y}. \tag{1}$$

The background effects such as scatters are not included, for simplicity. The solution of the liner system in (1) is conventionally found through the following constrained minimization problem:

$$\text{Minimize } \|\boldsymbol{x}\|_2^2/2 \text{ subject to } A\boldsymbol{x} = \boldsymbol{y}, \tag{2}$$

where $\|\boldsymbol{t}\|_2^2 = \sum_{i=1}^{n} t_j^2$ is the well-known ℓ_2 norm of the n-dimensional vector $\boldsymbol{t}$. When $m \ll n$, the linear system in (1) is underdetermined with infinite number of feasible solutions. Now, let us re-write this problem in a different form. Assume that the image $\boldsymbol{x}$ can be represented in a more sparse format using a sparsifying transform ψ. Such that:

$$|\psi(\boldsymbol{x})|_{\ell_1} \ll |\boldsymbol{x}|_{\ell_1}, \tag{3}$$

where $|\boldsymbol{t}|_{\ell_1} = \sum_{i=1}^{n} |t_i|$ is the ℓ_1 norm. The basic idea of CS is to reconstruct the image with a sparser domain transformation instead of the original non-sparse image. Thus, it is possible to reconstruct the small non-zero values from the down-sampled projection data. The non-sparse image can be obtained through an inverse transformation. However, in practice this step is not explicitly needed for the reconstruction. Instead of (2), the image reconstruction within the CS framework can be found by solving the following minimization problem:

$$\text{Minimize } |\psi(\boldsymbol{x})|_{\ell_1} \text{ subject to } A\boldsymbol{x} = \boldsymbol{y}. \tag{4}$$

This reformulation has a strong power in finding the solution when the problem is underdetermined. The reason is known as the "ℓ_1 magic" that define the effectiveness of ℓ_1 norm in finding the sparse solution compared to the conventional ℓ_2 norm. To demonstrate this important property of ℓ_1 norm, let us consider the following simple example. Assume that $n = 2$ and $m = 1$, which define a system of image consists of two pixels and a single projection measurement. The linear system to be solved is:

$$\Gamma\colon\ a_{11}x_1 + a_{12}x_2 = y_1. \tag{5}$$

To find the solution, the equi-contour lines corresponding to ℓ_1 and ℓ_2 norms are shown in **Fig. 1**. The intersection between the contour lines and the line that represent the linear system (5) is the solution of the problem. It is obvious from the illustration in **Fig. 1** that ℓ_1 norm is more likely to find the sparse solution (located at the coordinate

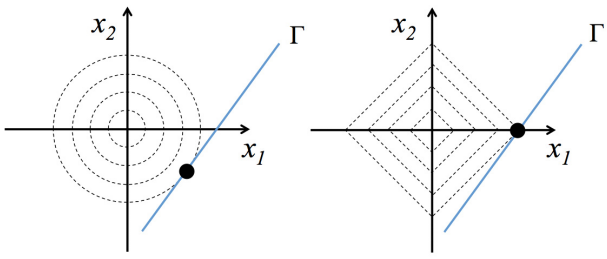

Fig. 1 Equi-contour lines (dashed) of the ℓ_2 (left) and ℓ_1 (right) norms. Γ is the straight line representing (5) and the solid point is the solution obtained through the minimization of the cost function.

axes) more than ℓ_2 norm. Obviously, this behavior is also valid for ℓ_0 norm. In real world problems, with significant higher-dimensions, the difference between ℓ_1 and ℓ_2 norms is significant.

In CT imaging, one measures the attenuation coefficients of x-ray photons within the scanned object. It is known that the variation of attenuation value within the same anatomy structure is rather small. Hence, it is not difficult to find a convenient specification transform. Simple gradient or wavelets transform can be used in most of diagnostic CT applications. Moreover, the Total-Variation (TV) minimization had become an essential useful tool that fit with this framework.

3. Reconstruction algorithms

In this section, a brief discussion on the recently developed image reconstruction algorithms and associated clinical applications in x-ray CT is discussed. For more details, we recommend the reader to cited references. Although, the CS framework is general that would fit with several data limitation problems, most of the efforts are focused on the problem of image reconstruction from a small number of projection views.

1）ASD-POCS algorithm

Total-variation norm was introduced for the first time as an efficient image denoising approach [4]. It is known for its edge preservation properties as the TV norm is based on non-quadratic function. TV minimization is being used as the standard approach for CS based reconstruction algorithms. The TV norm for two-dimensional function is computed as follows:

$$\|\boldsymbol{x}\|_{TV} = \sum_u \sum_v \|\nabla \boldsymbol{x}(u,v)\| = \sum_u \sum_v \sqrt{(\boldsymbol{x}(u+1,v) - \boldsymbol{x}(u,v))^2 + (\boldsymbol{x}(u,v+1) - \boldsymbol{x}(u,v))^2} \tag{6}$$

with u and v are the orthogonal Cartesian coordinates. The adaptive steepest-descent projection onto convex sets (ASD-POCS) algorithm is proposed as a constrained TV-minimization method for cone-beam CT [5]. The solution is obtained through the minimization of the following cost function:

$$\text{Minimize } \|\boldsymbol{x}\|_{TV} \text{ subject to } |A\boldsymbol{x} - \boldsymbol{y}| \le \epsilon,\ \boldsymbol{x} \ge 0 \tag{7}$$

where ϵ is a parameter that control data inconsistency. The gradient descent method is used to minimize the cost function, while a projection onto convex sets (POCS) method is used to enforce the data constraints. The ASD-POCS algorithm is implemented in an alternating manner between TV-minimization and data consistency enforcement. The same approach was earlier presented for data limitation problems in divergent-beam CT [6]. The ASD-POCS presents interesting reconstruction results from several data limitation problems, including limited views, limited angle, projection with bad detector bin, and limited views with bad detector bin [6]. It is also used to improve the image quality when being implemented to micro-CT real data acquired from low-dose acquisitions [7].

2）PICCS algorithm

In some CT applications, it is possible to obtain prior information about the scanned object in forms of reference image. The term reference image refers to an image with features that is very close to the target one. The reference image can be considered from an earlier scan of the same patient, or from analytical reconstruction from the union of interleaved dynamic projection in cardiac CT, for example. The prior image constrained compressed sensing (PICCS) algorithm is implemented such that the prior information from the reference image is incorporated in the image reconstruction process [8]. The PICCS algorithm is implemented by solving the following constrained minimization problem:

$$\text{Minimize } \left[\alpha|\psi_1(\boldsymbol{x} - \boldsymbol{x}_r)|_{\ell_1} + (1-\alpha)|\psi_2(\boldsymbol{x})|_{\ell_1}\right] \text{ subject to } A\boldsymbol{x} = \boldsymbol{y}, \tag{8}$$

where ψ_1 and ψ_2 are sparsify transforms, and $\boldsymbol{x}_r$ is the reference (prior) image. The PICSS algorithm was used to improve images reconstructed from few-views considering that a reference image is available. It is shown later that the PICSS algorithm is effective in improving the temporal resolution of four-dimensional cone-beam CT [9] and also useful in slow kVp switching acquisition in dual energy CT [10].

3）RARI algorithm

When the reconstruction process goes iterative, it is important to consider the computation cost. One conventional approach to speed up the reconstruction is to formulate the algorithm in the so-called row-action format. The image pixels update strategy in row-action models known to be very effective in reducing the number of required iterations to reach the optimum stage. The row-action reconstruction using a reference image (RARI) algorithm is developed within this framework [11]. The image reconstruction cost function is formulated as:

$$\text{Minimize } \|\boldsymbol{x}-\boldsymbol{x}_r\|_p^p/p \text{ subject to } A\boldsymbol{x}=\boldsymbol{y},\ \boldsymbol{x}\geq 0,\ (1<p\leq 2). \tag{9}$$

The minimization of the cost function in (9) is achieved through the implementation of the Lagrangian duality to convert the constrained minimization problem into an unconstrained maximization one. The final form of the reconstruction algorithm is based on a row-action image update. This method was implemented on multiphase abdominal CT toward the reduction of the patient dose. First, an initial contrast free image is acquired using normal scanning, and reconstructed image is then used in later contrast-based scans. A sample result of the RARI algorithm is shown in **Fig. 2**.

4）iMAP algorithm

The use of prior information obtained from a complete reference image is always being criticized to be inefficient in clinical use. In one hand, it is not always possible to obtain such a strong prior in many CT imaging applications. On the other hand, it is difficult to measure the contribution of the reference image in the target image. The aim of diagnostic CT is to acquire the true cross-sectional structure rather than a high-quality image. This motivates the work of intensity-based maximum *a posterior* (iMAP) algorithm [12]. In this algorithm, the prior information is limited to only a small number of intensity values that represent normal and expected structures (organs). The cost function is formulated to incorporate the intensity priors in the reconstruction and is defined as:

$$\begin{gathered}\text{Minimize } L(\boldsymbol{x})+\beta U(\boldsymbol{x}),\\ U(\boldsymbol{x})=\sum_{j=1}^{n}\min_l \omega_l\xi_l(x_j),\qquad \xi_l(t)=\begin{cases}|t-z_l|_{\ell_1} & z_{l-1}\leq t\leq z_l\\ \infty & (otherwise)\end{cases}\end{gathered} \tag{10}$$

where $L(\boldsymbol{x})$ is the negative log-likelihood function and $z=(z_1,z_2,\dots,z_L)$ is the known intensity values, and β is the parameter that controls the strength of the prior function $U(\boldsymbol{x})$. The iMAP algorithm is presented in a simple form that includes conventional iterative image update followed by a multi-thresholding function. The construction of the multi-thresholding function can vary based on the available intensity priors. The iMAP algorithm provides a potentially useful framework that can fit in several imaging applications and does not require a complete image as a reference. Reconstructed images from real head CT data using FBP and iMAP algorithms are shown in **Fig. 3**. The iMAP algorithm was extended later such that priors are computed from a probabilistic atlas [13]. This extension, not only useful in providing the necessary intensity priors, but also provides some spatial information in the image domain. The main idea is based on using previous scans of different patients to formulate a probabilistic atlas. This

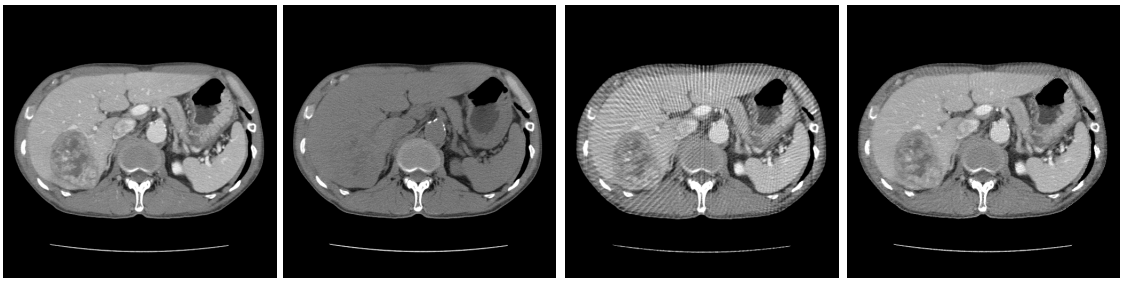

Fig. 2 From left to right, true image (obtained from full scan), reference image (obtained from contrast-free full scan), FBP and RARI reconstructions from limited views projections.

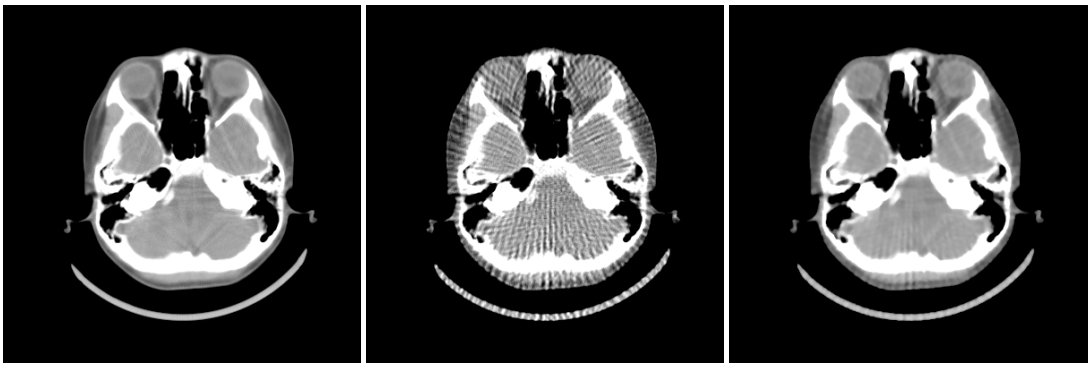

Fig. 3 True image obtained from full scan (left), FBP reconstruction from limited views (middle) and the iMAP reconstruction from the same limited data (right).

atlas is used to compute the intensity priors and how much each image pixel is likely being classified to specific organ.

There are several other interesting contributions to develop image reconstruction methods for x-ray CT; however, it is difficult to list everything in a single paper. We recommend the reader to the following reference include solutions to the interior problem [14], limited views [15, 16] and a comparison studies [17, 18].

4. Challenges and drawbacks

Recently, the concept of compressed sensing becomes a hot topic in several research fields. The early paper of Donoho [1] recorded more than 5,000 citations within only few years. However, the implementation in clinical applications is still not achieved. The main criticism is considering the mathematical background of the problem, which states that the exact reconstruction of down-sampled data is not theoretically possible. There is a possibility that some objects, which could be a lesion, are lost when the data measurements are not enough [19]. This is a serious drawback as the cost of providing an incorrect diagnosis is very high compared to the benefit of reducing the data measurements for the favor of patient dose reduction or other similar reasons. Considering the TV-minimization models, it is likely that TV-based algorithms perform very powerful through a simulated data, where the pixel intensity value within the same region is almost uniform. However, this is not always true when it goes to real data. It is known that TV-based reconstruction methods can produce artificial patches, known as carton-like artifacts, which reduce the possibility of being useful for diagnosis purposes [20]. As for now, it is still early to have a strict opinion regarding the impact of the CS based reconstruction algorithm in clinical CT applications.

5. Conclusions

Compressed sensing is becoming an interesting topic with wide range of applications, including x-ray CT. The significant improvement in image quality when projection data is limited provide a potential to be used in reducing the patient dose and provide a solution to conventional problems in x-ray tomographic imaging. This paper briefly discusses the main idea and introduces recent work in the field.

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